PROJECT REFERENCE / DATABASE DOCUMENTATION

HAWKER PROJECT DATA DICTIONARY

Singapore hawker-centre tender & location analytics
Reference version 3.0

A standalone reference to the MySQL **hawker_project** database. Historical NEA tender listings connect to official hawker centres, community stall observations, geographic boundaries, 2025 zoning, MRT/LRT access, station-level train ridership, SingStat June 2026 resident demographics and Google Maps planning-area cuisine engagement.

17	134	18	4
Physical tables	Physical fields	Foreign-key constraints	SQL views

THE SOURCE DATA AT A GLANCE

SOURCE	DATASET	HOW IT IS USED
1	**NEA Hawker Centres GeoJSON**	Official centre identity, postal address and location
2	**NEA tender notices (Oct 2025-Sep 2026)**	Historical cooked-food opportunities, AMR and charges
3	**Hawker Guide directory snapshot**	Community-observed occupancy, stall names, cuisine, prices and URLs
4	**URA Master Plan 2025 land use**	Zoned land-use composition within 500 m and 1 km buffers
5	**URA Master Plan 2019 subzones**	Planning geography, planning-area hierarchy and centre-to-subzone linkage
6a	**SingStat June 2026: dwelling type**	Residents by sex, age group and dwelling category
6b	**SingStat June 2026: floor area**	Residents by sex, age group and residential floor-area band
6c	**SingStat June 2026: single-year age**	Residents by individual age and sex; subzone totals
7	**LTA MRT/LRT station exits GeoJSON**	Rail exits, canonical station mapping, nearest access and 500 m counts
8	**LTA DataMall train ridership (Sep 2026)**	Station-level weekday and weekend/holiday passenger movements
9	**Google Maps reviews via Manus AI**	Planning-area cuisine engagement from scraped/crawled reviews

NOTES & DATA-QUALITY FLAGS

- Nine source groups are represented; Source 6 consists of three detailed June 2026 SingStat workbooks.
- Data cover different periods and measurement units. Resident subzones use MP2019 geography, land use uses MP2025, population counts refer to June 2026, and rail ridership covers September 2026.
- LTA dynamic train ridership is sourced from LTA DataMall (https://datamall.lta.gov.sg/content/datamall/en/dynamic-data.html); Google Maps review engagement was web-scraped/crawled using Manus AI.
- Planning-area cuisine engagement is shared by all hawker centres within the same planning area; it is not subzone- or centre-specific.

FIELD REFERENCE

Reading the tables

Each section identifies the **row grain**, main key and field definitions. PK and FK tags indicate SQL primary- or foreign-key roles. SQL views are documented after the physical tables. Examples are illustrative.

Physical database tables

01. hawker_centres

ONE ROW: One row per official NEA centre. KEY: centre_id. EXPECTED: 129.

COLUMN	MYSQL TYPE / KEY	EXAMPLE	DESCRIPTION
centre_id	*INT UNSIGNED* *NOT NULL* **PK**	3	Canonical centre identifier and preferred centre join key.
centre_key	*VARCHAR(160)* *NULL*	NULL / alias	Optional source-specific alias; NULL permitted. Join by centre_id.
nea_objectid	*BIGINT* *NULL*	119479	NEA GeoJSON feature ID; distinct from centre_id.
centre_name	*VARCHAR(255)* *NOT NULL*	Golden Mile Food Centre	Canonical display name of the hawker centre.
address	*VARCHAR(500)* *NULL*	505 Beach Road	Source hawker-centre street address.
postal_code	*CHAR(6)* *NULL*	048947	Six-digit Singapore postal code stored as text.
latitude	*DECIMAL(10,7)* *NULL*	1.3000000	Latitude in WGS84 decimal degrees (EPSG:4326).
longitude	*DECIMAL(10,7)* *NULL*	103.8600000	Longitude in WGS84 decimal degrees (EPSG:4326).
nea_status	*VARCHAR(120)* *NULL*	Existing	Latest administrative hawker-centre status in NEA source.
number_of_cooked_food_stalls	*SMALLINT UNSIGNED* *NULL*	100	Published NEA number of cooked-food stalls at a centre.
subzone_id	*INT UNSIGNED* *NULL* **FK**	1	2019 URA geographic subzone assigned to centre; FK to subzones.

02. centre_aliases

ONE ROW: One alias per source name and centre. KEY: alias_id. EXPECTED: Variable.

COLUMN	MYSQL TYPE / KEY	EXAMPLE	DESCRIPTION
alias_id	*INT UNSIGNED* *NOT NULL* **PK**	21	Internal surrogate identifier of a centre alias record.
centre_id	*INT UNSIGNED* *NOT NULL* **FK**	3	Project-assigned stable centre identifier, shared across data sources.
source_code	*VARCHAR(32)* *NOT NULL*	NEA_TENDER	Source identifier stored as text; no registry table or FK.
centre_name_in_source	*VARCHAR(255)* *NOT NULL*	Golden Mile Food Centre	Original centre name from a source, for name reconciliation.

03. stalls

ONE ROW: One identified stall per centre and unit. KEY: stall_id. EXPECTED: 284 (previous import).

COLUMN	MYSQL TYPE / KEY	EXAMPLE	DESCRIPTION
stall_id	*BIGINT UNSIGNED* *NOT NULL* **PK**	1	Project-assigned identifier of a physical stall (centre and unit number).
centre_id	*INT UNSIGNED* *NOT NULL* **FK**	3	Project-assigned stable centre identifier, shared across data sources.
stall_no_normalised	*VARCHAR(40)* *NOT NULL*	#02-80	Standardised stall unit within the centre; join with centre_id.

04. tender_opportunities

ONE ROW: One tender listing per stall and exercise month. KEY: listing_id. EXPECTED: 487.

COLUMN	MYSQL TYPE / KEY	EXAMPLE	DESCRIPTION
listing_id	*BIGINT UNSIGNED* *NOT NULL* **PK**	1	Unique tender appearance; one dashboard row per listing.
stall_id	*BIGINT UNSIGNED* *NOT NULL* **FK**	1	Project-assigned identifier of a physical stall (centre and unit number).
centre_id	*INT UNSIGNED* *NOT NULL* **FK**	3	Project-assigned stable centre identifier, shared across data sources.
exercise_month	*DATE* *NOT NULL*	2026-06-01	Month of NEA tender exercise; not current availability.
tender_open_date	*DATE* *NULL*	2026-06-12	Official date when the NEA monthly tender exercise opened.
tender_close_date	*DATE* *NULL*	2026-06-26	Official date when the NEA monthly tender exercise closed.
stall_no	*VARCHAR(40)* *NULL*	#02-80	Stall number as printed in tender notice.
release_category	*VARCHAR(40)* *NULL*	second_or_subsequent	Published NEA category: previously released no bids, first release, or subsequent release.
trade_marker	*VARCHAR(30)* *NULL*	+ / ^	Original symbols following stall number, for example + or ^.
allowed_trade_hint	*VARCHAR(255)* *NULL*	subject_to_pdf_notes	Broad interpretation of allowed trade; inspect official notes for definitive restrictions.
approximate_area_m2	*DECIMAL(9,2)* *NULL*	5.65	Approximate stall floor area in square metres from monthly PDF.
area_verified	*BOOLEAN* *NOT NULL*	1 (TRUE)	Whether the extracted stall area could be independently assigned in the PDF.
electrical_load_amps	*DECIMAL(8,2)* *NULL*	17.00	Approved electrical design load in amperes, if extracted and verified.
electrical_verified	*BOOLEAN* *NOT NULL*	1 (TRUE)	Whether the electrical load was verified at the correct stall in PDF.
amr_reference_year	*SMALLINT UNSIGNED* *NULL*	2026	Reference year for the displayed latest median assessed market rent.
median_centre_amr_sgd	*DECIMAL(12,2)* *NULL*	1,450.00	Centre median assessed market rent in SGD; not awarded rent.

COLUMN	MYSQL TYPE / KEY	EXAMPLE	DESCRIPTION
amr_2023_sgd	*DECIMAL(12,2)* *NULL*	1,450.00	Published centre-level median AMR for 2023 (SGD before GST), if available.
amr_2024_sgd	*DECIMAL(12,2)* *NULL*	1,450.00	Published centre-level median AMR for 2024 (SGD before GST), if available.
amr_2025_sgd	*DECIMAL(12,2)* *NULL*	1,450.00	Published centre-level median AMR for 2025 (SGD before GST), if available.
amr_2026_sgd	*DECIMAL(12,2)* *NULL*	1,450.00	Published centre-level median AMR for 2026 (SGD before GST), if available.
cleaning_fee_raw	*VARCHAR(200)* *NULL*	$200-$300	Exact cleaning-fee wording/amount extracted from monthly tender PDF.
cleaning_fee_kind	*VARCHAR(40)* *NULL*	range	Fee representation: range, greater-than limit, or special case.
cleaning_fee_min_sgd	*DECIMAL(12,2)* *NULL*	200.00	Lower endpoint when cleaning fee is a bounded range; not a quoted exact fee.
cleaning_fee_max_sgd	*DECIMAL(12,2)* *NULL*	300.00	Upper endpoint when cleaning fee is a bounded range; not a quoted exact fee.
cleaning_fee_strict_lower_bound_sgd	*DECIMAL(12,2)* *NULL*	300.00	Threshold when PDF states cleaning fee greater than a stated amount.
scc_raw	*VARCHAR(200)* *NULL*	$70-$90	Exact service and conservancy charge wording/amount from monthly tender PDF.
scc_kind	*VARCHAR(40)* *NULL*	range	Charge representation: range, greater-than limit, or special case.
scc_min_sgd	*DECIMAL(12,2)* *NULL*	70.00	Lower endpoint of published service and conservancy charge range.
scc_max_sgd	*DECIMAL(12,2)* *NULL*	90.00	Upper endpoint of published service and conservancy charge range.
scc_strict_lower_bound_sgd	*DECIMAL(12,2)* *NULL*	90.00	Threshold when published S&CC exceeds a stated amount.
costs_need_review	*BOOLEAN* *NOT NULL*	0 (FALSE)	Flag for charges requiring verification against source.
needs_review	*BOOLEAN* *NOT NULL*	0 (FALSE)	General flag requiring source verification.
raw_stall_token	*VARCHAR(120)* *NULL*	#02-80+	Original stall-number text including symbols from source PDF.
stall_note_context	*TEXT* *NULL*	PDF note 3	Published note references associated with the stall; do not assume restrictions from cuisine alone.
source_file	*VARCHAR(255)* *NOT NULL*	jun-2026-tender-notice.pdf	Exact monthly source PDF filename for tracing tender record.
source_page	*SMALLINT UNSIGNED* *NULL*	2	1-indexed PDF page number where tender listing appears.

AMR is a **reference rental value**, not a winning tender bid or current listing price.

05. stall_observations

ONE ROW: One imported Hawker Guide observation. KEY: observation_id. EXPECTED: 4,034.

COLUMN	MYSQL TYPE / KEY	EXAMPLE	DESCRIPTION
observation_id	*BIGINT UNSIGNED* *NOT NULL* **PK**	201	Unique surrogate key for each scraped stall observation.
centre_id	*INT UNSIGNED* *NOT NULL* **FK**	3	Project-assigned stable centre identifier, shared across data sources.
stall_id	*BIGINT UNSIGNED* *NULL* **FK**	1	Project-assigned identifier of a physical stall (centre and unit number).
observed_on	*DATE* *NULL*	NULL	Actual scrape/observation date, if independently recorded; SQL NULL for the current undated community scrape.
unit_no_raw	*VARCHAR(40)* *NULL*	02-80	Original Hawker Guide stall/unit number, including formatting.
unit_no_normalised	*VARCHAR(40)* *NULL*	#02-80 / NULL	Standardised unit; NULL may indicate a quarantined conflict.
occupancy_status	*VARCHAR(40)* *NULL*	occupied	Community directory-reported occupied/vacant state; not official NEA occupancy or tender availability.
store_name	*VARCHAR(255)* *NULL*	Example stall	Directory name of operating food stall, when observed.
cuisine_raw	*VARCHAR(150)* *NULL*	Chinese	Original scraped cuisine label.
cuisine_category	*VARCHAR(150)* *NULL*	Chinese	Standardized cuisine label used to count local competition (nine main-meal groups or Excluded - Non-main-meal; NULL if unknown).
price_range_raw	*VARCHAR(100)* *NULL*	$4-$8	Directory price range as reported; may not represent actual transaction prices.
postal_code_1	*CHAR(6)* *NULL*	048940	First postal code as contributed by Hawker Guide; not an authoritative location key. Use hawker_centres.postal_code for official values.
postal_code_2	*CHAR(6)* *NULL*	048947	Second postal code as contributed by Hawker Guide, if present; may disagree with the official NEA record.
source_url	*VARCHAR(2048)* *NOT NULL*	https://...	Original Hawker Guide web page URL for provenance.

The community snapshot is not a complete census of hawker-centre stalls. cuisine_raw is preserved; cuisine_category groups cuisines into nine main-meal groups plus Excluded - Non-main-meal (NULL remains unknown).

06. subzones

ONE ROW: One row per MP2019 subzone. KEY: subzone_id. EXPECTED: 332.

COLUMN	MYSQL TYPE / KEY	EXAMPLE	DESCRIPTION
subzone_id	*INT UNSIGNED* *NOT NULL* **PK**	1	Canonical MP2019 subzone ID; links centres and 2026 population.
planning_year	*SMALLINT UNSIGNED* *NOT NULL*	2019	Master Plan boundary vintage (2019).
subzone_code	*VARCHAR(30)* *NOT NULL*	BMSZ12	Subzone code as published by URA.

COLUMN	MYSQL TYPE / KEY	EXAMPLE	DESCRIPTION
subzone_name	*VARCHAR(150)* *NOT NULL*	DEPOT ROAD	URA MP2019 geographic label, not a guaranteed 2026 boundary.
planning_area_code	*VARCHAR(30)* *NOT NULL*	BM	Planning-area code as published by URA.
planning_area_name	*VARCHAR(150)* *NOT NULL*	BUKIT MERAH	URA MP2019 planning-area label; combined with subzone for matching 2026 SingStat records.
region_name	*VARCHAR(150)* *NULL*	CENTRAL REGION	URA planning-region name.
planning_area_id	*INT UNSIGNED* *NULL* **FK**	2	Links this subzone to planning_areas.planning_area_id (many subzones per planning area).

07. planning_areas

ONE ROW: One row per URA planning area. KEY: planning_area_id. EXPECTED: 55.

COLUMN	MYSQL TYPE / KEY	EXAMPLE	DESCRIPTION
planning_area_id	*INT UNSIGNED* *NOT NULL* **PK**	2	Canonical planning-area identifier; parent key for subzones and cuisine engagement.
planning_area_name	*VARCHAR(120)* *NOT NULL* **UNIQUE**	BUKIT MERAH	Normalised URA planning-area name.

08. population_2026_by_single_age

ONE ROW: One subzone/year/sex/age label. KEY: Composite 4-column key. EXPECTED: 91,632.

COLUMN	MYSQL TYPE / KEY	EXAMPLE	DESCRIPTION
subzone_id	*INT UNSIGNED* *NOT NULL* **PK/FK**	1	MP2019 subzone identifier; foreign key to subzones.subzone_id.
reference_year	*SMALLINT UNSIGNED* *NOT NULL* **PK**	2026	Reference year of the SingStat population release; 2026.
sex	*VARCHAR(8)* *NOT NULL* **PK**	Male	Standardised Total, Male or Female; original labels Males/Females normalised during R import.
age_label	*VARCHAR(30)* *NOT NULL* **PK**	25	Individual age as source label, or Total for all ages.
resident_count	*INT UNSIGNED* *NULL*	120 or NULL	2026 resident count by sex and individual age; dash is NULL.

Source dashes are stored as **NULL**; choose either total categories or component categories when aggregating.

09. population_2026_by_dwelling

ONE ROW: One subzone/year/sex/age band/dwelling type. KEY: Composite 5-column key. EXPECTED: 179,280.

COLUMN	MYSQL TYPE / KEY	EXAMPLE	DESCRIPTION
subzone_id	*INT UNSIGNED* *NOT NULL* **PK/FK**	1	MP2019 subzone identifier; foreign key to subzones.subzone_id.
reference_year	*SMALLINT UNSIGNED* *NOT NULL* **PK**	2026	Reference year of the SingStat population release; 2026.

COLUMN	MYSQL TYPE / KEY	EXAMPLE	DESCRIPTION
sex	*VARCHAR(8)* *NOT NULL* **PK**	Male	Standardised sex category: Total, Male or Female (sex-specific worksheets).
age_group	*VARCHAR(30)* *NOT NULL* **PK**	20-24	SingStat published age band, or Total for all ages; do not sum Total with component bands.
dwelling_type	*VARCHAR(100)* *NOT NULL* **PK**	4-Room Flats	Published category; totals and detailed categories overlap.
resident_count	*INT UNSIGNED* *NULL*	120 or NULL	2026 resident count by sex, age band and dwelling; dash is NULL.

Source dashes are stored as **NULL**; choose either total categories or component categories when aggregating.

10. population_2026_by_floor_area

ONE ROW: One subzone/year/sex/age band/floor band. KEY: Composite 5-column key. EXPECTED: 119,520.

COLUMN	MYSQL TYPE / KEY	EXAMPLE	DESCRIPTION
subzone_id	*INT UNSIGNED* *NOT NULL* **PK/FK**	1	MP2019 subzone identifier; foreign key to subzones.subzone_id.
reference_year	*SMALLINT UNSIGNED* *NOT NULL* **PK**	2026	Reference year of the SingStat population release; 2026.
sex	*VARCHAR(8)* *NOT NULL* **PK**	Male	Standardised sex category: Total, Male or Female (sex-specific worksheets).
age_group	*VARCHAR(30)* *NOT NULL* **PK**	20-24	Published age band, or Total for all ages.
floor_area_band	*VARCHAR(50)* *NOT NULL* **PK**	> 80 - 100	Published residential floor-area band (m²) or Total*.
resident_count	*INT UNSIGNED* *NULL*	120 or NULL	2026 count by sex, age band and floor area; dash is NULL.

Source dashes are stored as **NULL**; choose either total categories or component categories when aggregating.

11. centre_land_use_metrics

ONE ROW: One centre, plan year, buffer radius and land-use class. KEY: Composite 4-column key. EXPECTED: 4,079.

COLUMN	MYSQL TYPE / KEY	EXAMPLE	DESCRIPTION
centre_id	*INT UNSIGNED* *NOT NULL* **PK/FK**	3	Project-assigned stable centre identifier, shared across data sources.
plan_year	*SMALLINT UNSIGNED* *NOT NULL* **PK**	2025	Land-use plan vintage (2025 for project).
buffer_radius_m	*SMALLINT UNSIGNED* *NOT NULL* **PK**	500	Radius in metres used for geographical buffer around a hawker centre.
land_use_category	*VARCHAR(100)* *NOT NULL* **PK**	RESIDENTIAL	Normalized zoning class such as RESIDENTIAL or COMMERCIAL.
zoned_area_m2	*DECIMAL(16,2)* *NULL*	160,000.00	Mapped zoning area intersecting centre buffer, in m².

COLUMN	MYSQL TYPE / KEY	EXAMPLE	DESCRIPTION
land_use_share	*DECIMAL(9,6)* *NULL*	0.203718	Fraction of full radius buffer zoned to this class, not %. Zoning coverage may be below 100% near the sea; not footfall.

12. mrt_exits

ONE ROW: One mapped LTA rail exit. KEY: exit_id. EXPECTED: 613.

COLUMN	MYSQL TYPE / KEY	EXAMPLE	DESCRIPTION
exit_id	*INT UNSIGNED* *NOT NULL* **PK**	10	Internal unique ID of an LTA MRT or LRT station exit; referenced by centre_mrt_access.nearest_exit_id.
lta_objectid	*BIGINT* *NULL*	2822	External LTA OBJECTID for an MRT or LRT exit; not the internal primary key.
station_name	*VARCHAR(255)* *NOT NULL*	CC30	LTA station label; some entries use a code such as CC30.
exit_code	*VARCHAR(100)* *NULL*	Exit A	Published exit label, such as Exit 1, Exit A, etc.
rail_type	*VARCHAR(3)* *NULL*	MRT	Mode classification: MRT or LRT. Distinguishes transport modes for visual filters; does not imply relative passenger traffic.
latitude	*DECIMAL(10,7)* *NOT NULL*	1.3000000	Latitude in WGS84 decimal degrees (EPSG:4326).
longitude	*DECIMAL(10,7)* *NOT NULL*	103.8600000	Longitude in WGS84 decimal degrees (EPSG:4326).
station_id	*INT UNSIGNED* *NULL* **FK**	15	Canonical station ID; links exit to mrt_stations.station_id while preserving exit-level rail_type and geometry.

13. mrt_stations

ONE ROW: One canonical MRT/LRT station. KEY: station_id. EXPECTED: 184.

COLUMN	MYSQL TYPE / KEY	EXAMPLE	DESCRIPTION
station_id	*INT UNSIGNED* *NOT NULL* **PK**	15	Canonical station identifier shared by MRT and LRT exits.
canonical_station	*VARCHAR(255)* *NOT NULL* **UNIQUE**	STEVENS	Normalised station name without MRT/LRT STATION suffix.

14. mrt_station_codes

ONE ROW: One individual station code mapped to a canonical station. KEY: station_code. EXPECTED: Variable.

COLUMN	MYSQL TYPE / KEY	EXAMPLE	DESCRIPTION
station_code	*VARCHAR(40)* *NOT NULL* **PK**	TE11	Individual line/station code (e.g. DT10, TE11).
station_id	*INT UNSIGNED* *NOT NULL* **FK**	15	Canonical station for this code; multiple line codes at an interchange share station_id.

15. station_ridership

ONE ROW: One canonical station, reference month and day type. KEY: ridership_id (unique on station_id, reference_month, day_type). EXPECTED: 368.

COLUMN	MYSQL TYPE / KEY	EXAMPLE	DESCRIPTION
ridership_id	*BIGINT UNSIGNED* *NOT NULL* **PK**	1	Unique ridership record identifier.
station_id	*INT UNSIGNED* *NOT NULL* **FK**	15	Canonical station receiving the ridership observation; FK to mrt_stations.station_id.
source_pt_code	*VARCHAR(40)* *NOT NULL*	DT10/TE11	Original/approved LTA source code, possibly slash-combined for interchanges.
reference_month	*DATE* *NOT NULL*	2026-09-01	Reference month (first calendar day of month; e.g. 2026-09-01).
day_type	*VARCHAR(40)* *NOT NULL*	WEEKDAY	Day-type category: WEEKDAY or WEEKENDS/HOLIDAY.
pt_type	*VARCHAR(30)* *NULL*	TRAIN	Published transport type indicator from LTA source.
period_tap_ins	*BIGINT UNSIGNED* *NULL*	184,884	Total recorded passenger tap-ins during the reference period.
period_tap_outs	*BIGINT UNSIGNED* *NULL*	213,400	Total recorded passenger tap-outs during the reference period.
total_passenger_movements	*BIGINT UNSIGNED* *NULL*	398,284	Total station movements (period_tap_ins + period_tap_outs); not unique passengers.

Combined source identifiers such as DT10/TE11 represent one station-level record; individual codes remain in mrt_station_codes. The approved September 2026 study has 368 ridership rows.

16. centre_mrt_access

ONE ROW: One row per official hawker centre. KEY: centre_id. EXPECTED: 129.

COLUMN	MYSQL TYPE / KEY	EXAMPLE	DESCRIPTION
centre_id	*INT UNSIGNED* *NOT NULL* **PK/FK**	3	Project-assigned stable centre identifier, shared across data sources.
nearest_exit_id	*INT UNSIGNED* *NULL* **FK**	10	Nearest MRT or LRT exit after combining both transport types; FK to mrt_exits.exit_id.
nearest_exit_distance_m	*DECIMAL(12,2)* *NULL*	325.40	Straight-line metres to nearest MRT/LRT exit; not walking distance.
exits_within_500m	*INT UNSIGNED* *NULL*	3	Number of MRT and LRT exit points within 500 m.
stations_within_500m	*INT UNSIGNED* *NULL*	1	Number of distinct MRT/LRT stations with exits within 500 m.

17. cuisine_engagement_by_planning_area

ONE ROW: One planning area and standardised cuisine group. KEY: engagement_id (unique on planning_area_id, cuisine_group). EXPECTED: Variable.

COLUMN	MYSQL TYPE / KEY	EXAMPLE	DESCRIPTION
engagement_id	*INT UNSIGNED* *NOT NULL* **PK**	1	Unique planning-area cuisine engagement row identifier.

COLUMN	MYSQL TYPE / KEY	EXAMPLE	DESCRIPTION
planning_area_id	*INT UNSIGNED* *NOT NULL* **FK**	2	Planning-area reference; FK to planning_areas.planning_area_id.
cuisine_group	*VARCHAR(100)* *NOT NULL*	Chinese	Standardised main-meal cuisine group (nine groups).
restaurant_count	*INT UNSIGNED* *NULL*	48	Count of sampled Google Maps restaurants in this cuisine group and planning area.
total_reviews	*BIGINT UNSIGNED* *NULL*	16,402	Total Google Maps reviews across sampled restaurants in this cuisine and planning area.
avg_reviews	*DECIMAL(14,2)* *NULL*	341.71	Mean reviews per sampled restaurant; may be NULL if no restaurants are sampled.
engagement_share	*DECIMAL(18,15)* *NULL*	0.1941	Share of total sampled review volume within the planning area (fraction 0 to 1, not %).

The nine cuisine groups measure planning-area-level Google Maps review engagement, not actual stall sales, demand, or centre-level footfall.

ANALYTICAL VIEWS

SQL views

Views calculate results from base tables and contain no separate stored population copy. Always check the output grain before joining them.

v_subzone_population_2026

Grain: one row per MP2019 subzone (332 rows); 2026 totals selected from the appropriate detailed source categories.

OUTPUT FIELD	SOURCE / MEANING
subzone_id	Primary geographic join key; one row per MP2019 subzone.
reference_year	Reference year of all three SingStat datasets.
resident_population	June 2026 resident total. NULL if unpublished for subzone.
male_residents	Male resident total; NULL if no numerical count.
female_residents	Female resident total; NULL if no numerical count.
hdb_population	All-ages, both-sexes HDB residents; a subset of total resident population.
floor_area_covered_population	Resident count covered by the floor-area source universe; not additive to resident_population.

Total selection: single-age sex=Total, age_label=Total; dwelling sex=Total, age_group=Total, dwelling_type=Total HDB^; floor area sex=Total, age_group=Total, floor_area_band=Total*.

v_tender_dashboard

Grain: one row per tender listing (487 unique listing_id values). Recommended starting point for tender-level maps and filters.

OUTPUT FIELD	SOURCE / MEANING
listing_id	Unique tender listing identifier; primary reporting grain.
exercise_month	Monthly tender exercise reference date.
tender_open_date	Date tender opened, if reported.
tender_close_date	Date tender closed, if reported.
centre_id	Official linked hawker-centre key.
centre_name	Canonical centre name.
postal_code	Official centre postal code as text.
latitude	Centre WGS84 latitude.
longitude	Centre WGS84 longitude.
stall_id	Linked tender stall identifier.
stall_no_normalised	Normalised tender stall number.
release_category	Tender release category.
trade_marker	Published symbols associated with stall number.
allowed_trade_hint	Broad trade interpretation; consult source restrictions.
approximate_area_m2	Approximate stall area in square metres.
electrical_load_amps	Approved electrical design load, amperes.
median_centre_amr_sgd	Centre assessed market rent reference (SGD); not awarded rent.
amr_reference_year	Reference year of the stated centre AMR.
cleaning_fee_kind	Representation of fee (range/threshold/other).
cleaning_fee_min_sgd	Cleaning fee lower endpoint, when applicable.
cleaning_fee_max_sgd	Cleaning fee upper endpoint, when applicable.

OUTPUT FIELD	SOURCE / MEANING
scc_kind	Service/conservancy charge representation.
scc_min_sgd	S&CC lower endpoint.
scc_max_sgd	S&CC upper endpoint.
needs_review	Tender listing quality-review flag.
costs_need_review	Fee/charge review flag.
subzone_name	2019 matched subzone name.
planning_area_name	2019 matched planning-area name.
region_name	URA planning-region name.
resident_population	2026 total resident population, where reported.
hdb_population	2026 HDB resident total, where reported.
nearest_exit_distance_m	Straight-line metres to nearest combined MRT/LRT exit.
exits_within_500m	MRT+LRT exit points within 500 m.
stations_within_500m	Unique rail stations with exits within 500 m.
nearest_mrt_station	Legacy name: nearest station may be MRT or LRT.
residential_share_500m	Share of full 500 m buffer zoned RESIDENTIAL.
commercial_share_500m	Share of full 500 m buffer zoned COMMERCIAL.
population_reference_year	2026 for matched subzones; NULL where no geography joins.
male_residents	Male residents, subzone level, June 2026.
female_residents	Female residents, subzone level, June 2026.
floor_area_covered_population	Covered floor-area population, not additive.
nearest_rail_type	Nearest exit mode: MRT or LRT.

Use for stall-level maps and filters; do not join raw many-row cuisine or ridership tables directly without pre-aggregating or filtering.

v_cuisine_mix

Grain: one row per centre and classified cuisine in the cleaned community directory.

OUTPUT FIELD	SOURCE / MEANING
centre_id	Hawker-centre identifier.
cuisine_category	Normalised main-meal cuisine category (excludes NULL and Excluded - Non-main-meal).
observed_stalls	Number of latest occupied and classified unit observations for cuisine.
share_of_classified_occupied_stalls	Proportion among classified, occupied observed stalls in that centre.

Excludes NULL and Excluded - Non-main-meal categories so shares sum across classified main-meal stalls per centre.

v_centre_cuisine_engagement

Grain: one row per centre_id and cuisine_group when mapped via subzone and planning area.

OUTPUT FIELD	SOURCE / MEANING
centre_id	Hawker-centre identifier.
centre_name	Canonical display name of the hawker centre.
subzone_id	Linked MP2019 subzone identifier.
planning_area_id	Canonical URA planning-area identifier.

OUTPUT FIELD	SOURCE / MEANING
planning_area_name	Normalised URA planning-area name.
cuisine_group	Standardised main-meal cuisine group.
restaurant_count	Sampled Google Maps restaurants in this cuisine group within the centre's planning area.
total_reviews	Total Google Maps reviews for sampled restaurants in the planning area.
avg_reviews	Mean Google Maps reviews per sampled restaurant in the planning area; may be NULL.
engagement_share	Share of planning-area Google Maps review volume (0 to 1); replicated across centres in the same planning area.

Joins hawker_centres -> subzones -> planning_areas -> cuisine_engagement_by_planning_area. All centres in the same planning area share identical engagement metrics.

DATABASE RELATIONSHIPS

Keys & safe joins

The 18 enforced foreign-key relationships below define the v3.0 post-cleanup schema. A two-column relationship (stall_id, centre_id) must be joined using **both** fields.

CHILD TABLE / KEY	PARENT TABLE / KEY
hawker_centres (subzone_id)	subzones (subzone_id)
subzones (planning_area_id)	planning_areas (planning_area_id)
centre_aliases (centre_id)	hawker_centres (centre_id)
stalls (centre_id)	hawker_centres (centre_id)
tender_opportunities (stall_id, centre_id)	stalls (stall_id, centre_id)
tender_opportunities (centre_id)	hawker_centres (centre_id)
stall_observations (centre_id)	hawker_centres (centre_id)
stall_observations (stall_id, centre_id)	stalls (stall_id, centre_id)
centre_land_use_metrics (centre_id)	hawker_centres (centre_id)
centre_mrt_access (centre_id)	hawker_centres (centre_id)
centre_mrt_access (nearest_exit_id)	mrt_exits (exit_id)
mrt_exits (station_id)	mrt_stations (station_id)
mrt_station_codes (station_id)	mrt_stations (station_id)
station_ridership (station_id)	mrt_stations (station_id)
population_2026_by_single_age (subzone_id)	subzones (subzone_id)
population_2026_by_dwelling (subzone_id)	subzones (subzone_id)
population_2026_by_floor_area (subzone_id)	subzones (subzone_id)
cuisine_engagement_by_planning_area (planning_area_id)	planning_areas (planning_area_id)

QUICK JOIN MAP

QUESTION	RECOMMENDED PATH
Tenders	tender_opportunities -> hawker_centres via centre_id; -> stalls via stall_id and centre_id (or use v_tender_dashboard; preserve listing_id uniqueness).
Population	hawker_centres -> subzones -> v_subzone_population_2026 using subzone_id (one subzone summary; do not sum across tender rows).
Age / housing charts	subzones -> one filtered population_2026_by_* table, not all three raw detail tables at once.
Rail access	hawker_centres -> centre_mrt_access -> mrt_exits (nearest_exit_id to exit_id).
Station volume	mrt_exits.station_id -> mrt_stations.station_id -> station_ridership.station_id; filter reference_month and day_type to avoid repeating volume per exit.
Zoning	hawker_centres -> centre_land_use_metrics; filter plan_year and buffer_radius_m.
Cuisine supply	hawker_centres -> v_cuisine_mix on centre_id; filter selected cuisine or pre-aggregate to centre.
Cuisine engagement	hawker_centres -> v_centre_cuisine_engagement on centre_id (+ cuisine_group), or subzones.planning_area_id -> planning_areas -> cuisine_engagement_by_planning_area.

INTERPRETATION RULES

- Do not sum subzone population across tender rows: centres and tenders can repeat the same subzone.
- Do not sum published Total categories with their component age, sex, dwelling or floor-area categories.
- 2026 subzone matching uses MP2019 geographic references; matching names do not prove unchanged boundaries.
- MRT/LRT access is straight-line proximity. Land-use zoning is not observed commercial activity or footfall.
- Missing population is NULL, not zero. Only 281 of 332 subzones have a published numeric total in the single-age source.
- NEA median AMR is a reference rent, not the awarded bid or a currently available rental offer. Tender listings cover Oct 2025-Sep 2026.
- Rail station ridership measures period tap-ins + tap-outs, not unique people, daily averages or traffic at a specific exit/hawker centre. Keep WEEKDAY and WEEKENDS/HOLIDAY separate.
- Google Maps cuisine engagement measures share of sampled review counts within a planning area, not actual cuisine demand or revenue. When joining one-to-many tables, aggregate or filter to the desired grain first.